# exoChile2026: Chilean Exoplanet Community Meeting

## April 27-29, 2026, UCSC Concepción, Chile

*Last updated: April 27, 2026*

## Day 1 Morning

8:30 — **Registration and arrivals.**

9:15 — **Welcome & Opening Remarks**.

Overview of objectives (*Matías Jones*) and format of the meeting (*Seba Pérez*)

9:30 — **Keynote** (*chair: Matías Jones*)

Andrés Jordán *Overview of Exoplanet Research from Chile*

10:10 — **Key Community Updates**. Short updates (7 min each) from different groups (*chair: Matías Jones*)

- David R. Anderson *La búsqueda de exoplanetas en tránsito liderada por la UCN*
- James Jenkins *Exoplanet research from UDP/CATA*
- Rafael Brahm *The WINE collaboration: Characterizing warm jupiters to solve the hot Jupiter puzzle*
- Jeremy Tregloan-Reed *Exoplanets in Chile International collaborations and northern state telescopes*
- Alice Zurlo *Exoplanet research from YEMS*
- *Gijs Mulders / Claudia Aguilera* *Exoplanets at PUC*
- Open time for updates

11:10 — **Coffee break**

11:40 — **"Vaciado": Community Mapping.** Each (sub-)community or groups identifies:

Current strengths — Tools, facilities, and datasets — Ongoing/planned projects — Gaps, needs, opportunities and dreams

12:45 — **Group presentations**

## Day 1 Afternoon

14:30 — **Lightning Talks (3 min per talk)** *(chair: Máx Moyano)*

- H. Parashivamurthy TESS analysis of super-Earths around super small stars
- Ricardo Ramírez Time-domain Hα spectroscopy of wide-orbit planetary-mass companions with VLT/MUSE
- Michaela Vitkova Warm gas giants and their friends
- Scarlett E Royle The effect of stellar position-velocity clustering and galactic dynamics on exoplanet properties
- Ignacia Santiago Long-Term Dynamical Stability of Multiplanet Systems' Rocky Exoplanets in the Habitable Zone
- Andrea Bernardi Spectrophotometric analysis and orbital monitoring of directly imaged substellar companions
- Alejandra Fernández Autoencoders for Speckle Denoising in ADI Sequences
- Joanne Rojas Molina RV-Astrometry Synergy in the Detection of Warm Giants
- Ilaria Giovannini Looking for satellites with doppler measurements of directly imaged exoplanets and brown dwarfs
- Bárbara Rojas-Ayala Bayesian Model Averaging for Homogeneous Stellar and Exoplanet Characterisation from Late-K to Ultracool Dwarfs

16:00 — **Coffee break**

16:30 — **Contributed talks (10 min each)** *(chair: Nicola Astudillo-Defru)*

- Felipe Lagos-Vilches Observational Constraints on the Incidence of Substellar Companions to Young Massive White Dwarfs with VLT/ERIS
- Carolina Charalambous Lithium abundance in TOI-xxx: testing the planet-engulfment hypothesis
- Connor Cheverall Ground-based Atmospheric Characterization of Volcanic Super-Earth L98-59d at High Spectral Resolution
- Adam Rains VLT/CRIRES+ observations of warm Neptune WASP-107 b: Molecular detections and challenges in ground-based transmission spectroscopy of cooler and cloudy exoplanets
- Kevin Hoy First Evidence of Radial Velocity Signal from Satellite(s) Around a Substellar Companion
- Gijs Mulders Exoplanet Water Worlds: What are they and where can we find them?
- Claudia Aguilera-Gómez Polluted White Dwarfs as Probes of Exoplanetary Systems
- Jhon Yana Galarza What Can Alpha-Enhanced Stars Tell Us About Exoplanets?
- Marcela Best Kepler in Kepler: how an "alien" astronomer would develop Kepler's Third Law

## Day 1 Evening

18:00 — **Community Icebreaker** (food/drinks)

**19:00 — Public talk "Exoplanetas y vida en el espacio" by Alice Zurlo at Universidad Católica de la Santísima Concepción**

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## Day 2 Morning

9:30 — **Group presentations from Community Mappings. [NEW]**

10:30 — **Panel discussion [NEW]**

- AI for exoplanets — Connor Cheverall, Cecilia Hernández and James Jenkins

11:15 — **Coffee Break**

11:45 — **Mixed-Community Dialogues.** Participants rearrange into **diverse/mixed groups**, focusing on 1–2 questions, for example:

- Collaboration opportunities
- Chile's role in global exoplanet science
- Barriers to collaboration

13:00 — Lunch break

## Day 2 Afternoon

14:30 — **Lightning talks (3 min each)** *(chair: Alice Zurlo)*

- Fernando Castillo M. — Survival of vortices in gas-rich debris disks
- Erick Montoya — Unveiling the real geometry of the FU Orionis system
- Constanza Norambuena Herrera — Polluting white dwarfs: a dynamical study
- Alex Mauch-Soriano — Predicted incidence of Jupiter-like planets around white dwarfs
- Sidharth Aroli Veettil — The 14N/15N and 12C/13C isotopic ratios in the protoplanetary disk around V4046 Sgr
- Constanza Castillo — The formation of formaldehyde (H2CO) in protoplanetary disks
- Marie M. Rodríguez S. — Variability and Geometry of the Brγ Emission in HD 100546 from VLTI/GRAVITY
- Milenne Avila Bravo — Planetary architectures under the influence of a stellar binary
- Lucas Gonzalez-Rivas — Not Just Gas: How Solid-Driven Torques Shaped the Migration of the Galilean Moons
- Grace Batalla Falcon — Evolution of the silicate feature in protoplanetary disks
- Camilo González-Ruilova — Can the environment affect the planet formation?
- Hannah Osborne — A homogenous analysis of small-planet masses

15:10 — **Contributed talks (10 min each)** *(chair: Gijs Mulders)*

- Nicola Astudillo-Defru — Searching for terrestrial exoplanets with optimized radial velocities
- Claudio Caceres — A search for free-floating planets in nearby young stellar associations
- Neda Hejazi — Exploring Planetary Formation and Composition Using Detailed Chemical Abundances of Host Stars
- Philipp Weber — Planet formation in the light of stellar accretion bursts
- Matthias Schreiber — The importance of tides for the survival of planets during late stellar evolution

16:00 — **Coffee Break**

16:30 — **Mixed-communities Report-Back (30 min)**

- Groups summarize key ideas
- Emergent themes guide the seed-lab

17:00 — **Collaboration Seed-Labs.** Self-organized breakout groups form around concrete ideas. For example:

- Joint proposals
- Shared data/AI tools or open repositories
- Training programs or student exchanges
- Outreach and public engagement initiativ

18:00 — [tbc] **Community Strategy Session** (Roadmap Discussion). Possibly as a small panel

- National priorities
- Future facilities
- Student training pathways

## Day 2 Evening

**19:00 — Public talk "Del Desierto de Atacama al cosmos" by Bárbara Rojas-Ayala at Universidad Católica de la Santísima Concepción**

## Day 3 Morning

10:00 — **Seed-Lab Pitches (5 min/each).** Groups present concrete next steps and timelines.

10:30 — **Student-led reflection session.** A session *organized by students, moderated by postdocs.*

- Training needs
- Mentoring networks
- What barriers do students face in Chile?
- Community structure (how can the community better support early-career researchers?)
- Guided reflections on the need of national summer schools on exoplanet research, statistics and machine learning, or other topics
- Chilean exoplanet Slack workspace

11:00 — **Coffee Break**

11:30 — **From ideas to Working Groups. Working Groups and Commitments.** Here's where we consolidate initiatives from seed labs and other sessions.

- Organizers compile “exoChile 2026 Community Initiatives”

12:00 — **Community Roadmap Discussion**

12:30 — **Next Steps and Closing Remarks**

- Who hosts exoChile2027 (or 2028?)
- Creation of communication channels

13:00 — **Farewell and Group Photo**

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## List of Posters

| | | |
|---|---|---|
| **ATM-01** | Benjamin Gonzalez Gonzalez Vidal | Modeling of Archean Earth-like aquaplanet atmosphere: the effect of obliquity on climate and observables of exoplanets |
| **ATM-02** | Catalina Valesca Ávalos Vega | Breaking Degeneracies in Exoplanet Atmospheres: Measuring Rayleigh Slopes in exoplanets to Infer Mean Molecular Weight |
| **DET-01** | Sebastián Elias Cabrera Astudillo | BODIES: Big Or Dinky Investigation and Evaluation System |
| **DET-02** | Marcela Cristina López Ulloa | When M Dwarfs Flicker: Are Young Exoplanets Hiding There? |
| **DET-03** | Daniela Gonzalez | The Synergy of Gaia DR4 and VLTI/GRAVITY+: Low-Mass exoplanet detection in Binary Systems |
| **DET-04** | Joanne Michelle Rojas Molina | RV-Astrometry Synergy in the Detection of Warm Giants |
| **DET-05** | Florian Rodler | Mass determination of the three long-period Neptune- and sub-Neptune-sized planets transiting TOI-282 |
| **DET-06** | Cathalina Andrea Zamora Fernández | Discovery and characterization of a long period system |
| **DET-07** | Franshezka Dominik Paz Tapia Candia | The discovery and characterization of a long period system |
| **DET-08** | José Luis Ricra Mayorca | Orbital period refinement and analysis of central transit time variations of WASP-50 b using ground-based and space observations |
| **DET-09** | Jose Vines | TIC 275647095 b, an eccentric Warm Jupiter Orbiting a Solar-analog Star |
| **DET-10** | Maximiliano Moyano | Discovery and characterization of a long period system |
| **DISK-01** | Hernán René Salinas Castro | Searching for Complex Organic Molecules with JWST–MIRI in Planetesimal Formation Regions |
| **DISK-02** | Abigali Rodríguez Jiménez | Rotation curves of prototype protoplanetary discs |
| **EDU-01** | Sol Francisca Troncoso Barrías;<br>Lindsay Rincón Torres;<br>Thiare Antonella Molina Tapia | Introduction to Data Science through the Exploration of Exoplanets as an Educational Strategy |
| **INST-01** | Oscar Castillo Vega | An optimized implementation of the PACO algorithm. |
| **INST-02** | Ian Rickmers Blamey | Deepermoments: Deep Learning for moment maps |
| **OUT-01** | Martina Pascall Molina Riffo;<br>Bárbara Scarlett Quilodrán Hijerra;<br>Javiera Cajas Reyes | Astronomical Collage as a Science Communication Tool with a Gender Perspective in Exoplanet Science Outreach |
| **SYS-01** | Felipe Ignacio Rojas Henríquez | Giant companions in Super-Earth systems from TESS |
| **SYS-02** | Álvaro Ignacio Llancaqueo Albornoz | Unveiling the nature of TESS giant planets in the first PLATO Field |
| **SYS-03** | Bhavya Ambalapatta kannancheri | Is the Hot Neptune Desert Really Empty? A Comparison of TESS and Kepler. |

**Poster Code Legend** (Numbers indicate individual posters within each thematic category.)

DET — Detection & Characterization of Exoplanets
ATM — Exoplanet Atmospheres & Climate
SYS — Planetary Systems & Demographics
DISK — Protoplanetary Disks & Planet Formation
INST — Instrumentation, Methods & Algorithms
DS — Data Science, Machine Learning & Education
OUT — Outreach, Communication & Society

# Your Role as Facilitator

You are not just moderating; you are **enabling/fostering collaboration**. Your responsibilities:

- Keep discussions **focused and on time**
- Ensure **everyone participates briefly**
- Identify **connections between participants**
- [Desirable but not compulsory] Push groups toward some **concrete outcomes**

# General Principles (apply throughout all sessions)

- Prioritize **clarity over completeness**
- Avoid long monologues, better to have **short interventions**
- Actively ask: **"What do you need?" and/or** What do you need to be happy/productive? Reach your goals? What are your goals?
- Look for:
    - Shared datasets or tools
    - Complementary expertise
    - Overlapping scientific questions
- Push toward **smaller groups** when natural connections appear
- [Desirable but not compulsory] Steer toward **something actionable**

# Day 1 — Community Mapping ("Vaciado", 11:40, duration 1hr)

We will work in different **"mesas de trabajo" / Work groups**. LOC will communicate locations to facilitators via Slack.
Groups are identified by the **color of the sticker** on your name badge.

## Step 1 — [very important] Rapid Introductions (suggested duration ~20 min)

Each participant briefly states:

- Name & institution
- Research focus
- What they are currently working on
- **What they need / are looking for** (this is key).

Keep this **short (~1-2 minute per person)**

## Step 2 — Live Mapping (~15 min)

As people speak, the facilitator should try to **capture patterns**, such as:

- Common research themes
- Shared datasets or instruments
- Similar methods
- Complementary needs

## Step 3 — Trigger Connections (~15 min)

Use guiding questions:

- Who needs data/expertise that someone else has?
- Who is working on similar problems?
- Are there theory–observation complementarities?
- Could this become a paper or proposal?

## Step 4 — [desirable but not compulsory] Form Small Clusters

Encourage spontaneous **subgroups (2–4 people)** to emerge around:

- Shared interests
- Complementary skills
- Common goals

**Output of this session:** A **clear map of the community. And hopefully** initial **clusters of potential collaboration**

# Day 2 — Mixed-Community Dialogues (10:30, duration 1 hr)

Now participants are **re-mixed across communities**.

### Your goal here: expand and challenge ideas, activate student participation if it doesn't happen naturally (ex: by asking what they know about other subfields or what interests them most).

Guide discussion around:

- Collaboration opportunities across subfields
- Chile's role in global exoplanet science
- Barriers (technical, institutional, cultural)

### Push beyond discussion, and don't forget to GRAB A COFFEE (coffee break actually *breaks* the session in two)

Encourage participants to:

- Connect ideas from Day 1 clusters
- Identify **new interdisciplinary opportunities**
- Refine or rethink collaboration ideas

**Key facilitator action:** If promising ideas emerge: Encourage those people to **stay connected for the afternoon Seed-Labs**

# Day 2 — Seed-Labs (15:40, duration ~40 min)

This is where ideas *could* become **concrete collaborations**. Facilitators can circulate between Seed-Lab groups, acting as catalysts to help those ideas become collaborations.

## Step 1 — Form Groups Around Ideas

Participants self-organize into groups based on:

- Ideas from Day 1 or morning discussions
- New collaborations emerging during the meeting

## Step 2 — Guide Toward a Concrete Plan

Each group should define (there will be a printed template/form to complete with this info):

- **What are we doing?** (clear problem/topic)
- **Who is involved?**
- **Why this collaboration?**
- **What is the concrete outcome?** Paper? Proposal? Shared dataset/tool? Training initiative?

## Step 3 — Define First Steps

Push groups to specify:

- First actions
- Rough timeline
- Communication method

If there is no next step, it is not a real collaboration.

# Day 2 — Seed-Lab Pitches (short summaries) (16:45)

Help groups prepare a **clear, concise pitch (5 min max)**:

They should answer:

- What is the idea?
- Who is involved?
- What will you produce?
- What are the next steps?

Encourage clarity and realism over ambition.